

Ganesh Yandigeri

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SUMMARY

Python engineer building LLM/RAG applications and low-latency backend systems with FastAPI and React. Built a full-stack RAG platform on Google Gemini and ChromaDB, and a matching engine processing 136,000 orders/sec with price-time priority and pre-trade risk checks.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL

Backend & Systems: FastAPI, Flask, Node.js, REST APIs, WebSockets, AsyncIO, Microservices, Low-Latency System Design, Data Structures & Algorithms

AI/ML: RAG Pipelines, Vector Search (ChromaDB), Google Gemini API, Semantic Embeddings, Prompt Engineering

Databases: PostgreSQL, MongoDB, Redis

Frontend: React.js, Next.js, Tailwind CSS

Tools & Infra: Docker, Git, Linux, CI/CD, Pytest

PROJECTS

Florix AI — Full-Stack AI Academic Engine & RAG Platform Jan 2025 – Present
Python, FastAPI, Google Gemini API, ChromaDB, SQLAlchemy, React 19, Tailwind CSS

- Built a full-stack academic engine that ingests multi-format sources — PDFs, YouTube transcripts, web articles — and generates adaptive quizzes and active-recall flashcards using Google Gemini
- Built a RAG pipeline on ChromaDB (HNSW index) with metadata filtering for semantic retrieval, measured under 200 ms per query, with SQLite fallback for local development
- Built an asynchronous REST API secured with JWT auth, role-based authorization, and SlowAPI rate limiting; modeled study streaks and user telemetry across 15+ relational tables via SQLAlchemy

OrderCore — Limit Order Book & Matching Engine Sep 2026
Python, FastAPI, AsyncIO, WebSockets, Pytest

- Built a single-symbol matching engine with strict price-time priority, supporting LIMIT, MARKET, IOC, and FOK order types, pre-trade risk checks, and self-trade prevention
- Designed the order book using a heap-indexed price-level structure with $O(1)$ amortised order cancellation, achieving 136,000 orders/sec throughput and $4.3\mu\text{s}$ median match latency on a single core
- Exposed a REST order-entry API and a WebSocket market data feed streaming live order book depth to connected clients in real time
- Validated matching correctness with a 14-test suite covering price/time priority, partial fills, self-trade prevention, and cash conservation invariants

EXPERIENCE

Software Development Intern Feb 2025
Dharwad-Hubballi Tutor / Remote

- Optimized React frontend components, improving page load speed by 20%; resolved critical authentication bugs securing 100+ active users
- Integrated RESTful APIs enabling real-time data fetching and dynamic content rendering

EDUCATION

Atria Institute of Technology — Bangalore Expected May 2027
Master of Computer Applications (MCA)

Basaveshwara Science College — Bagalkot 2022 – 2025
Bachelor of Computer Applications (BCA) | CGPA: 8.3/10

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Generative AI Professional — Oracle University, Oct 2025
- Oracle Cloud Infrastructure 2025 AI Foundations Associate — Oracle University, Aug 2025
- Crash Course on Python — Google, Coursera, Sep 2025